DroneVPS - Project Status

Marker-free GNSS-denied navigation on a 4.0 kg hexacopter. Target: 100-150 m AGL, 3-5 km out-and-back, no GNSS, no ground marker. Two flights since the last report, 246 and 247. Flight 247 flew a full waypoint mission on the VPS from takeoff and touched every waypoint - a first - then lost its hold below 60 m and took an EKF failsafe at 377 s.

Overall

79%

78% on 25 October. The rise is the first autonomous mission and the altitude fault finally being measured rather than argued. Nothing new has flown, so nothing new is above 95%.

The barometer is 5.5 m high, and now it is proven.

Against the GPS altitude on log 246, which is independent of both: lidar = 0.898*GPSalt - 0.04 m at corr 0.980, baro = 0.889*GPSalt +5.49 m at corr 0.989. Same slope - that is the GPS - and the offsets are the finding. Velocity is pixels x altitude / focal, so +5.5 m is +7% of velocity at 100 m and +70% at 10 m.

The map cannot work below 60 m with this sheet.

276 frames at 28-62 m, four window floors - 40 m as flown, 25, 15 and none - accepted 0 of 276 in every one. The query is rescaled down to the map's 0.509 m/px, so at 30 m it is 56x56 px and XFeat finds 9 matches against a gate of 12. At 80 m it is 150x150 px and 145.

The lidar's zero wanders metres, and the node now handles it.

6.94 m read with the lens 0.26 m above the floor; 0.65 -> 2.9 m in 30 minutes untouched. Not the supply: Vcc sd 0.0058 V, corr with the reading -0.110. The scale is right. The node re-learns the zero on the ground until the moment of arming.

The handover is clean. The glitch is not the handover.

Measured on both switches into the VPS on log 246: the first injected position was 0.01 m from what the EKF held, after dropping 31 m and 33 m of accumulated drift. The GPS-glitch and EKF messages arrive 0.4 to 3.4 s LATER.

An item is 100% only when it has been verified in flight.

1. Platform and plumbing

Item		Status / evidence
Airframe, compute, sensors	**95%**	ZD850 hexa, CUAV X7 Pro, Jetson Orin, OAK-D Pro. The pilot re-measured the stack: FC 6 cm above the CG, camera 22 cm below. GPS2_POS_Z corrected 0.14 -> 0.22 on the aircraft.
Camera and frame rate	**98%**	Unchanged. Both monos at 25 fps, CAM_C primary, no crop.
Vibration	**97%**	Re-measured across five flights at the pilot's challenge: X 0.89-1.35, Y 0.91-1.37, Z 1.37-1.95 m/s2 median, zero clipping on all of them. ArduPilot's limit is 30. Not a fault and removed from the list.
MAVLink routing	**94%**	Unchanged. Parameters were set live through the router today - rangefinder zero and GPS2_POS_Z - with the node running and its rangefinder stream intact afterwards.

Services and auto-start	**100%**	Unchanged.
Flight data recording	**96%**	80-column CSV, frames at 4 Hz. Down from 97: flight 247 produced NO node-side CSV at all, so that flight can only be read from the FC log.
Preflight diagnostics	**94%**	The lidar zero no longer needs a preflight command - the node measures it itself while disarmed and on the ground.

2. Position estimation

Item		Status / evidence
Optical flow	**92%**	Unchanged. Per-track derotation since 25 October.
Dead reckoning over distance	**85%**	Unchanged method.
Gyro derotation	**90%**	Unchanged. The model is right; the gain never was the problem.
Camera / IMU configuration	**95%**	Unchanged.
Mount alignment	**90%**	Camera yaw offset -1.95 deg. The ROLL mount offset is still unset while pitch is -1.13; roll MAD was 2.43 at 110 m, which is +-4.6 m of scatter.
Map matching	**78%**	Down from 80, on knowledge rather than performance. Above 60 m it is unchanged and good - 5.64 m p50, 2.89 m at 60-80. Below 60 m it is now known to be impossible with this sheet, so lowering map_min_alt_m is closed. A finer map is the only route left, and only for low altitude.
Sensor fusion	**72%**	Unchanged. fix_sigma_m is still 20 while the map now measures 5.64 m, so the Kalman gain sits at 0.068. The correction written for this was reverted and has not been brought back.
Velocity to the FC	**55%**	At 40-60 m on flight 246 the sent speed was 2.03 m/s where the receiver said 0.94. The barometric offset explains about 0.09 of that 1.09 m/s. The rest is the phantom and is still open.
Heading / yaw	**55%**	Unchanged. The compass is still the only yaw source.
Altitude	**60%**	Up from 45. The fault is measured and a fix is written and unflown: the lidar corrects the barometer's offset without replacing it, and the lidar's own zero is re-learned on the ground to the moment of arming. EK3_SRC1_POSZ stays 1 and EK3_RNG_USE_HGT stays -1 - this sensor is fit to measure a slow offset, not to carry height.
Position injection as GPS2	**92%**	GPS2_POS_Z now 0.22 m, the camera's true offset below the CG.

3. Flight controller integration

Item		Status / evidence
EKF origin and home	**100%**	From SCR_USER1/2 at arm on both flights, which is what a jammed environment needs. The pilot has closed the question of taking it from the receiver and is right to.

Manual RC GPS1/GPS2 switch	**100%**	Measured on both switches into the VPS on log 246: 0.01 m from the EKF's own position, 31 m and 33 m of drift dropped. Nothing to fix.
Actually flying on the VPS	**80%**	Down from 85. Flight 246 held - 3.95 m at 40-60 m against the receiver. Flight 247 did not: 27.23 m at 40-60 and 24.83 at 60-80, and an EKF failsafe to ALT_HOLD at 377 s.
Smoothness	**35%**	Unchanged and still the lowest line on the sheet. The deadband is a hard edge; the ramp written for it is in history and not brought back.
Autonomous mission	**70%**	Up from 40. Flight 247 flew the whole waypoint mission on the VPS from takeoff and touched every waypoint. It lost the hold on descent, which is why this is not higher.
Mission speed ceiling	**70%**	Unchanged.
Behaviour on RC loss	**100%**	Unchanged.

4. What moved since 25 October

Up

Autonomous mission	40% ->	**70%**	247 flew every waypoint on the VPS
Altitude	45% ->	**60%**	+5.5 m measured against GPS altitude; fix written
Vibration	96% ->	**97%**	re-measured on five flights, zero clipping
Preflight diagnostics	92% ->	**94%**	the lidar zeroes itself now
Position injection as GPS2	90% ->	**92%**	GPS2_POS_Z 0.14 -> 0.22

Down

Actually flying on the VPS	85% ->	**80%**	247 took an EKF failsafe below 60 m
Map matching	80% ->	**78%**	below 60 m is now known to be impossible here
Flight data recording	97% ->	**96%**	247 produced no node CSV at all

5. Corrections - all of these are mine

I called the 40 m window floor the direct answer to the descent problem.
It was arithmetic. Measured on 276 real frames, all four window floors accepted 0 of 276. The fault is the query's pixel count, not the window.

I measured a 43 m step at the handover into the VPS.
My own artefact: injection stops while the switch is on the receiver, so the nearest GPS2 sample was minutes stale. The real step is 0.01 m.

I told the pilot to apply the 20-25 September rangefinder configuration.
Applied and measured: the reading went 6.94 m -> 19.72 m. Restored. Lower scalings were then swept at the pilot's instruction - 65, 50, 30, 10 - and all of them shrink the reading rather than zero it.

I started to doubt that the barometer reads high.
The GPS-altitude check confirmed the original finding. The premise the altitude fix rests on holds.

I put vibration on the to-do list without measuring it first.
The pilot asked why it had never been reported. Measured: fine on every flight. Removed.

I wrote a three-command field procedure for the lidar zero.
The pilot rejected it on design, not convenience, and was right. The node does it now and there is nothing to run.

6. Open, in the order I would take them

1. Velocity at 40-60 m	2.03 m/s sent where the receiver said 0.94. The barometric offset explains 0.09 of it. This is the largest open fault.
2. Smoothness	307 zero crossings in 634 s. The deadband ramp is written and waiting on the pilot, because it changes a guard.
3. The lidar's zero	4.5 m/hour at a fixed distance today. Not the supply. A two-distance test an hour apart separates a drifting sensor zero from a drifting resistor; only the second is fixed by a 25 ppm/C metal-film part.
4. Roll mount offset	Unset. Pitch is -1.13. Roll MAD 2.43 at 110 m.
5. fix_sigma_m 20	The map now measures 5.64 m. The gain is 0.068.
6. A finer map, below 60 m only	The only route left to map fixes during the descent. Above 60 m a finer sheet gives nothing - 9.0 m against 8.9.